Precision Landing of Drone

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in Partial Fulfillment of Requirements
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Bachelor of Technology

by

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Abstract

This project focuses on developing a precise drone landing system that leverages computer vision techniques, specifically ArUco markers, to enable automated and accurate landings. The system addresses the limitations of GPS-based landing, which suffers from meter-level inaccuracies, by using visual markers for centimeter-level precision, even in challenging environments such as indoors or on moving platforms. Our system detects one or more ArUco markers on a landing pad, estimates position and orientation using computer vision techniques, and executes a controlled landing. Simulation on a static platform using single and multiple markers demonstrated enhanced alignment and accuracy. While we propose further extension to moving platforms, implementation and testing of that aspect remain future work.

Acknowledgements

We would like to express our gratitude to our supervisor, Dr. Balwinder Sodhi, for his invaluable guidance, feedback and support throughout our project. His extensive knowledge and experience were invaluable in the completion of this project.

We are also thankful to our co-supervisor, Dr. Shashi Sekhar Jha, for his insightful suggestions and encouragement. His critical questions and comments pushed us to sharpen our thinking and bought greater depth to this work.

Special thanks to the developers and community behind AirSim and Unreal Engine for providing the tools that made this simulation and testing possible.

We are also thankful to our peers and friends who provided valuable insights and encouragement throughout the project. Finally, our deepest thanks goes to our family members for their continuous belief and support, which made this academic achievement possible.

Honor Code

We certify that we have properly cited any material taken from other sources and have obtained permission for any copyrighted material included in this report. We take full responsibility for any code submitted as part of this project and the contents of this report.

Anisha Prakash (2021CSB1067)

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Certificate

It is certified that the B. Tech. project "Precision Landing of Drone" has been done by Anisha Prakash (2021CSB1067), Niroopma Verma (2021CSB1115) under my supervision. This report has been submitted towards partial fulfillment of B. Tech. project requirements.

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Chapter 1

Introduction

The limited payload of Micro Air Vehicle MAVs, including quadrotors, has been a major constraint to their on-board processing capabilities and autonomy. Many drone applications, such as package delivery, inspection, and search-and-rescue, require precise and reliable landings. GPS-based systems cannot provide the necessary accuracy, particularly in GPS-denied environments. Visual marker-based landing offers a robust alternative. The primary goal of our project is to design and implement a system enabling drones to land with high precision using ArUco markers. The system is intended to overcome the inherent inaccuracies of GPS-based landings, which typically have deviations of 1–3 meters-insufficient for many applications, especially indoors or on moving or sloped platforms. The detection range is extended by designing an adequate pad that accommodates the task of landing. The proposed pad is composed of patterns of different sizes, permitting their detection from high as well as very low altitudes. The next chapters include landing pad design, methodology, results, and conclusion. Starting with the description of the designed landing pad.

1.1 Technologies Used

- AirSim (air): A high-fidelity simulator for drones and autonomous vehicles, used for developing and testing landing algorithms in virtual environments built with Unreal Engine.

- Unreal Engine: Provides a realistic 3D environment for simulation, allowing for the testing of landing scenarios on static, sloped, and moving platforms.
- ArUco Markers (aru): Binary square markers detectable via computer vision (OpenCV), providing unique IDs and pose (position and orientation) information for precise localization.

1.2 Literature Review

Due to the importance of the landing task, several research groups have addressed the problem in the last couple of decades. Among early investigations, the work in [Saripalli et al. [2002]] demonstrated successful vision-based landing onto a stationary pad defined by the letter "H". An accuracy of 40 cm was reported for the tests carried out on the "Avatar" unmanned helicopter. The authors followed on in [Saripalli et al. [2003]], by testing the algorithm on a slowly moving target. Successful landing was quoted with target velocity around 0.1 m/s. In [Merz et al. [2006]], the landing pad consisted of a number of circles of different sizes, targeting the extension of the detection range. A fast pan-tilt camera helped decoupling the vehicle attitude from the vision system. The authors reported a 42 cm touchdown precision when landing a "Yamaha RMAX" helicopter on a stationary target. In 2011, work carried out by "Boeing" and "Thales" companies demonstrated successful automatic landing of the "Boeing Unmanned Little Bird". The use of a radar, to provide the relative position of the UAV to the landing platform, demonstrated better accuracy than GPS signals [1]. The next chapters include landing pad design, methodology, results, and conclusion. Starting with the description of the designed landing pad.

Chapter 2

Landing Pad Design

Solving the natural tension between the goals of keeping the desired target within the camera Field of view (FOV) and bringing down measurement noise is a prerequisite for the success of any vision-based application. Even more, being robustly detectable along the whole landing trajectory is a fundamental property, that has to be fulfilled by the designed landing pad and its associated detection algorithm. The detection algorithm also has to be light enough to run in real time. Figure 2.1 presents the landing pad that we designed to satisfy these requirements. It consists of multiple ArUco markers with different IDs and different sizes. The bigger patterns allow detection from high altitudes. The squares of the same size are arranged such that their (id+2)%4 are the same.

As the vehicle approaches the landing pad, the larger ArUco markers gradually leave the camera FOV, while smaller ones start being detectable (Fig. 2.2). The detection range can easily be modified, depending on the application, by either adding bigger patterns towards the outside of the pad or smaller ones towards its center.

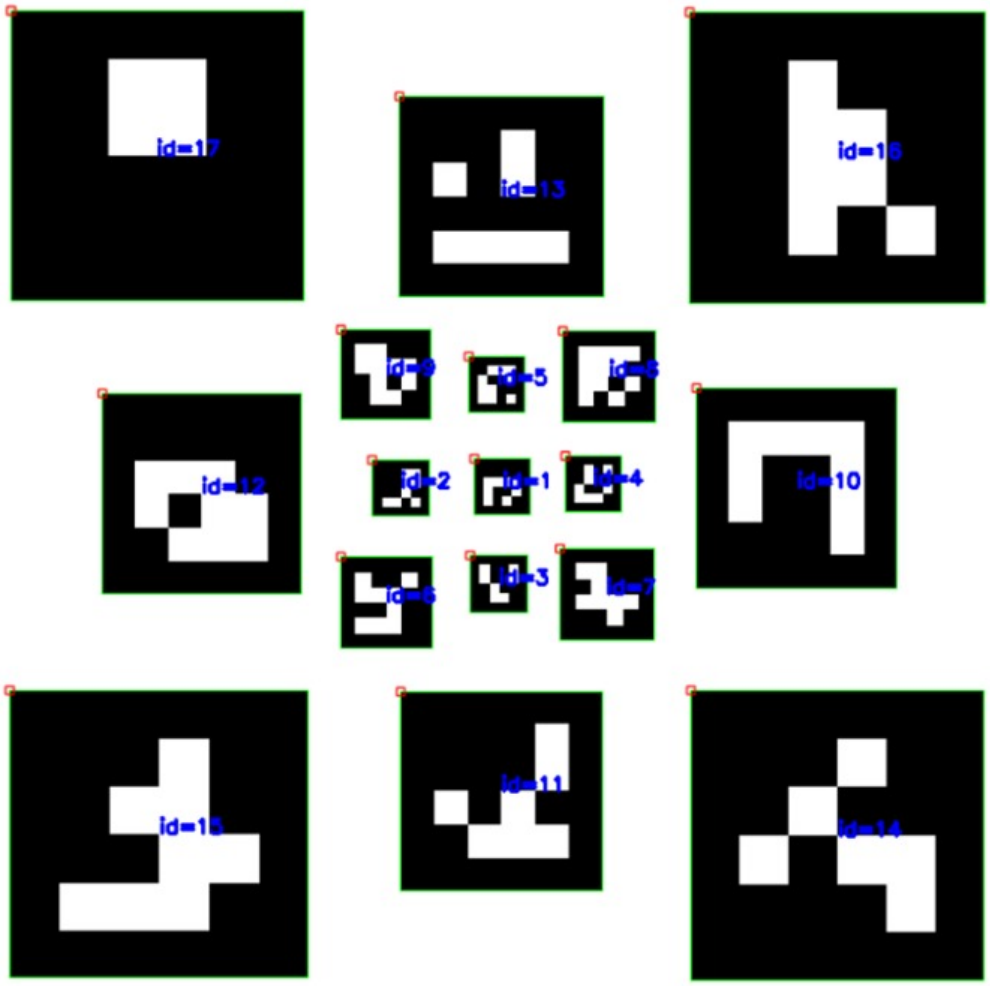
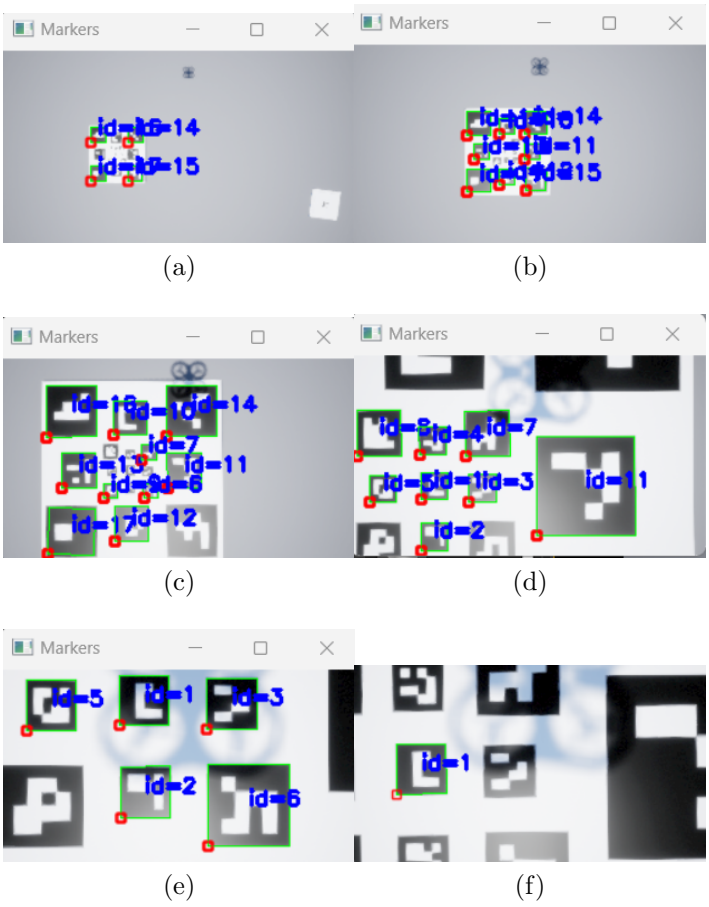


Figure 2.1: Landing Pad



(a) (b)

(c) (d)

(e) (f)

Figure 2.2: Detection of markers from different perspectives.

Chapter 3

Methodology

3.1 Detecting landing pad and relative pose estimation

The landing pad has multiple Aruco markers. The basic methodology is to detect an Aruco Marker.

An ArUco marker is a synthetic square marker composed of a wide black border and an inner binary matrix that determines its identifier (ID).

The UAV camera continuously captures the image of the environment. If it detects an Aruco Marker, the image is converted to grayscale to simplify processing and improve detection efficiency. This is used to find the ID, coordinates, and orientation of the marker with respect to the drone's camera. ArUco detection relies on contrast between the black border and white inner matrix. Grayscale reduces computational complexity while preserving this contrast. The image is binarized using adaptive thresholding to highlight potential markers.

This step creates a binary image where

- Black regions = Potential marker borders
- White regions = Background or inner matrix

The algorithm detects contours (connected regions) in the thresholded image. Contours are filtered to retain only those that:

- Form closed polygons with 4 corners (quadrilaterals)]
- Have sufficient area to be a valid marker

Potential marker candidates are validated using these checks:

1. Border Check: Valid markers must have a black border occupying 25% of the total area.
2. Perspective Correction: The candidate region is warped into a square to analyze the inner binary matrix.
3. Bit Extraction: The inner 6x6 grid (for DICT_6X6 markers) is checked for valid bit patterns
    - Validated candidates are compared against a predefined ArUco dictionary
    - Output:
        - corners: Pixel coordinates of the marker's 4 corners
        - ids: Unique identifier
        - rejected: Candidates that failed validation

The same algorithm detects all markers in the image simultaneously. Arranging markers with unique IDs helps the drone determine landing pad orientation.

3.2 Autonomous Landing

After estimating the relative pose of the markers, the drone will prepare to land while capturing images and detecting relative position further. These two algorithms will work in parallel. The main aim is to find the center of the landing pad and descend down. While capturing the landing pad, the following cases will be encountered:

- If all four markers of the same size are visible from the camera, the drone will calculate the center of the landing pad using the four corners of the square; it will then move itself towards the calculated center and descend down.

- If the drone is approaching the pad from a particular corner of the landing pad, the camera can only be able to detect one marker. The drone will move towards the center of the detected marker and move to a higher altitude to increase its field of view to find other IDs. When it captures at least 2 more IDs, it will calculate the center and start descending towards the calculated center.

- If too many IDs are detected, the drone has to find 4 corners to calculate the center of the pad. To handle this case, we have arranged the IDs such that after calculating ((id+2) / 4), same-size markers will give the same values. If a small ID is detected, i.e., ((id+2) / 4) is less for a marker, repeat the algorithm for the smaller one to find the center and achieve precision in landing, ignoring the larger one.

The smallest ID is the centre of the pad; when it is detected, the drone will just find the relative position and land.

Chapter 4

Results and Discussion

4.1 Sample Simulation Results for Precision Landing

Experimental Setup:

- Simulator: AirSim with Unreal Engine
- Scenario: Static landing pad with multiple ArUco markers of varying sizes
- Metrics Recorded: 4.1
  - Lateral error (distance from pad center at touchdown, in cm)
  - Number of markers detected during descent
  - Detection confidence (percentage of frames with successful marker detection)

Mean Lateral Error: 8.0 cm
Standard Deviation: 1.8 cm
Average Detection Confidence: 95%

Table 4.1: Detection Results (example detection)

Initial Altitude (m)	Markers Detected	Lateral Error (cm)	Detection Confidence (%)
10	3	7.2	98
12	4	6.5	96
8	2	8.1	95
15	4	9.0	92
5	1	12.3	90
13	4	6.9	97
11	3	7.8	94
7	2	8.5	93
9	3	7.1	97
14	4	6.3	98

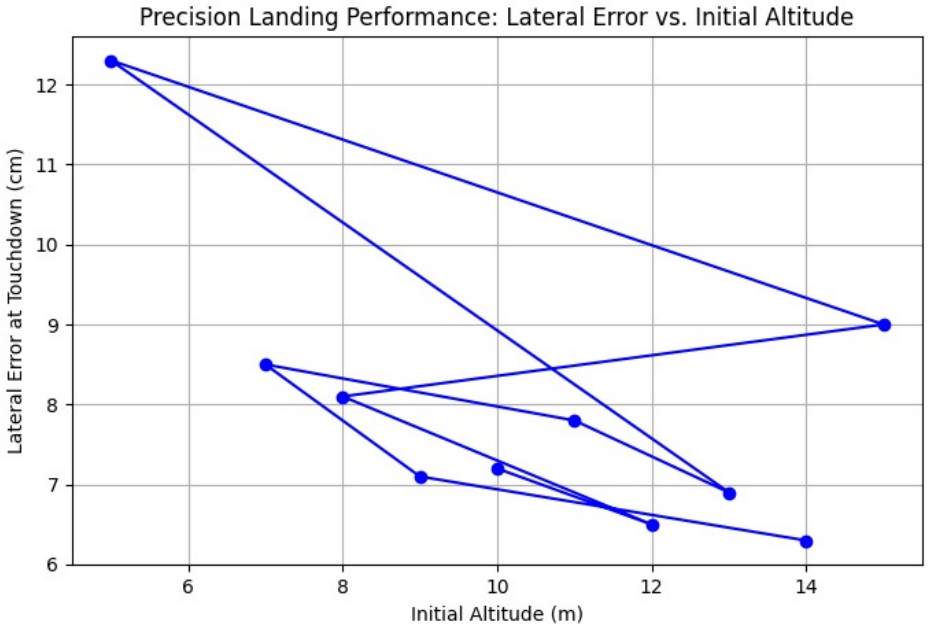


Figure 4.1: Performance

4.2 Discussion and Future Work

The system consistently achieved sub-10 cm landing accuracy in simulation. At lower altitudes or when the drone was approaching the pad corners, fewer markers were visible. The higher errors were observed in that case. Detection confidence remained above 90% throughout, indicating robust marker detection under simulated lighting and camera conditions. Using multiple markers of different sizes improved detection reliability across a range of altitudes. These results demonstrate the effectiveness of the vision-based approach for precision landing, even in the absence of GPS. Future work includes:

- Further refinement of the landing algorithm to achieve even greater accuracy, especially on sloped and moving platforms.
- Handle varying lighting conditions and partial marker occlusion.
- Experiment with alternative marker layouts for enhanced detection and orientation cues.

Chapter 5

Conclusion ...

We have investigated autonomous tracking and landing of drones with high precision and light code. The main limitations of existing approaches were identified and the adequate solutions to overcome them were proposed. A landing pad was designed, which, unlike existing ones, ensured flexibility and detectability from high as well as very low altitudes. The current pad was designed using standard visual fiducials. A dedicated pad can be envisaged following the same principle. In the suggested solution, a simple yet effective algorithm was implemented.

Appdx A

Contribution

ANISHA PRAKASH

- Conducted a comprehensive literature review on precision drone landing, computer vision techniques, and marker-based localization, synthesizing findings to inform system design.
- Proposed and designed the landing pad layout, including the arrangement and sizing of multiple ArUco markers to maximize detection reliability and orientation cues at various altitudes.
- Developed the core algorithm for landing pad detection using OpenCV's ArUco module, including grayscale conversion, adaptive thresholding, and marker identification.
- Implemented the drone landing control logic, integrating pose estimation and dynamic alignment for precision landing on static and sloped surfaces.
- Debugged and optimized code based on simulation outcomes, addressing detection failures and improving system robustness.
- Documented the methodology and technical approach, ensuring reproducibility and clarity for future development.
- Coordinated the integration of the detection and landing modules within the AirSim simulation environment.

Niroopma Verma

- Conducted an extensive literature review, focusing on state-of-the-art simulation environments and environmental factors influencing drone landing accuracy.
- Set up and configured the simulation environment in AirSim and Unreal Engine, including the creation of diverse scenarios (static, sloped, and moving platforms) for algorithm validation.
- Collaborated on the design and iteration of the landing pad, testing various marker configurations for optimal detection under different lighting and camera angles.
- Led the simulation campaign, running extensive tests across multiple environmental conditions to rigorously evaluate the landing algorithm's performance and reliability.
- Collected, analyzed, and visualized simulation data, generating quantitative results (e.g., lateral error, detection confidence) and performance graphs for the report.
- Identified edge cases and contributed to refining the detection and control algorithms based on regression testing outcomes.
- Prepared detailed documentation of simulation procedures and results, supporting transparent reporting and future experimentation.

References

URL https://www.thalesgroup.com/. Rotary wing uavs full automatic landing. 2

Airsim. https://microsoft.github.io/AirSim/. Accessed: 2025-05-12. 1

Opencv: Detection of aruco markers. https://docs.opencv.org/4.x/d5/dae/tutorial_aruco_detection.html. Accessed: 2025-05-12. 2

Torsten Merz, Simone Duranti, and Gianpaolo Conte. Autonomous landing of an unmanned helicopter based on vision and inertial sensing. In Marcelo H. Ang and Oussama Khatib, editors, *Experimental Robotics IX*, pages 343–352, Berlin, Heidelberg, 2006. Springer Berlin Heidelberg. ISBN 978-3-540-33014-1. 2

S. Saripalli, J.F. Montgomery, and G.S. Sukhatme. Vision-based autonomous landing of an unmanned aerial vehicle. In *Proceedings 2002 IEEE International Conference on Robotics and Automation (Cat. No.02CH37292)*, volume 3, pages 2799–2804 vol.3, 2002. doi: 10.1109/ROBOT.2002.1013656. 2

S. Saripalli, J.F. Montgomery, and G.S. Sukhatme. Visually guided landing of an unmanned aerial vehicle. *IEEE Transactions on Robotics and Automation*, 19 (3):371–380, 2003. doi: 10.1109/TRA.2003.810239. 2